

Unit – 5 : Session – 6 : SLO – 1 | Symbolic Calculus — Differentiation and Integration**I. Questions & Answers**

1. Differentiate x^3 — $d/dx(x^3) = 3x^2$. In SymPy: `diff(x**3, x) → 3*x**2`.
2. Integrate $2x$ — $\int 2x \, dx = x^2 + C$. In SymPy: `integrate(2*x, x) → x**2` (constant omitted).
3. Differentiate $\sin(x)$ — $d/dx \sin(x) = \cos(x)$. In SymPy: `diff(sin(x), x) → cos(x)`.

II. Code

1. `from sympy import symbols, diff, integrate, sin`
2. `x = symbols('x')`
3. `print(diff(x**3, x)) # 3*x**2`
4. `print(integrate(2*x, x)) # x**2`
5. `print(diff(sin(x), x)) # cos(x)`